

Shashank Kamble

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Summary

Master's student in IT & Analytics at Rutgers University focused on RAG systems, LLM evaluation, and NLP applications. Experience building and benchmarking retrieval pipelines, conversational AI evaluation frameworks, and multilingual NLP systems using Python and modern AI tooling. Interested in fast-moving AI and data-driven engineering roles.

Education

Rutgers University

Aug 2025 – Present

M.S. IT & Analytics (AI Concentration) | GPA: 3.53

Relevant Coursework: LLMs, Machine Learning, Business Analytics, Data Management, Data Visualization.

St. Francis Institute of Technology

Jun 2021 – May 2025

B.E. Computer Science (Cyber Security Honors) | CGPA: 8.57/10

Skills

Programming: Python, SQL, R, C

Machine Learning: Regression, Classification, Clustering, Model Evaluation, Feature Engineering, LLM Fine-Tuning

NLP: Text Preprocessing, Sentiment Analysis, Multilingual NLP, Transformer Fine-Tuning, Feature Extraction

Data Analysis: Exploratory Data Analysis (EDA), Data Cleaning, Statistical Analysis

Frameworks & Libraries: Pandas, NumPy, Scikit-learn, HuggingFace Transformers, OpenCV, MediaPipe

Deployment & Tools: Docker, Vercel, Git/GitHub, Tableau, Power BI, Excel, Cursor, Claude Code, GitHub Copilot

Experience

All India Radio (AIR) | *Research & Programming Intern*

Jun 2023 – Jul 2023

- Worked with broadcast and audience datasets to analyze engagement trends and support programming insights
- Streamlined reporting workflows using Python and Excel for faster analysis and cleaner operational tracking

CodSoft | *Data Analytics Intern*

Jun 2024 – Jul 2024

- Developed analytical dashboards and KPI reports using Python and BI tools to visualize trends across structured datasets
- Automated preprocessing and exploratory analysis workflows, improving consistency and reducing repetitive manual analysis

Rutgers Institutional Planning & Operations | *Office Assistant, Mail Services*

Feb 2026 – Present

- Managed high-volume logistics and maintained structured tracking records in a fast-paced operational environment.

Projects

RAG-based Housing Recommendation System (NJ)

- Benchmarked 5 retrieval pipelines (BM25, Dense, Hybrid, Query Rewriting, SPLADE-v3 + Reranker) across 7 NJ public-service domains using a manually curated 70-question evaluation benchmark
- Improved Recall@5 from 0.52 (baseline dense retrieval) to 0.69 using SPLADE-v3 with Qwen3-Reranker-4B, achieving the strongest retrieval performance across all evaluated methods
- Reduced hallucination rate to 2.86% while achieving 76.43% answer accuracy through grounded generation with Qwen2.5-32B-Instruct
- Built category-aware FAISS and SPLADE retrieval indexes over 46K+ structured chunks, optimizing retrieval quality versus latency trade-offs for deployment-oriented RAG workflows

LLM Response Evaluation System for Conversational AI (Therapeutic Domain)

- Fine-tuned and evaluated 5 open-source LLMs on a 600-sample psychotherapy dataset spanning 13 therapeutic techniques
- Built multi-axis evaluation framework combining SBERT similarity, sentiment analysis, and human validation to benchmark conversational quality
- Identified Falcon and Gemini models as strongest performers for therapeutic coherence and contextual consistency through embedding-based evaluation and PCA analysis

FactCheckr – Multilingual Fake News Detection (NLP)

- Built multilingual fake-news detection pipelines for Hindi and English datasets using NLP preprocessing, feature extraction, and machine learning techniques
- Achieved 99.3% classification accuracy using Gradient Boosting models, outperforming Logistic Regression and Random Forest baselines

Extracurriculars

Computer Society of India (CSI) | *Creative Head*

Jul 2022 – May 2024

- Led creative direction and coordinated cross-functional team of 25 students for technical and non-technical events